Maximum Likelihood Estimation Examples

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Key Concepts

1) Maximum likelihood estimation seeks to find the parameter that maximizes the probability density function evaluated at the observed data.

2) Closed form solutions for the maximum likelihood estimate can often be obtained in problems with simple (e.g., Gaussian) probability density functions.

Video Screenshots

Maximum likelihood – $\hat{\underline{\alpha}} = \arg\max_{\underline{\alpha}} \{ f(\underline{x}; \underline{\alpha}) \}$

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Example 1: Mean and Variance

Suppose $x[n] \sim N(\mu, \sigma^2)$, $n = 0, 1, \ldots, N-1$ independent

Estimate μ, σ^2 from $x[0], x[1], \ldots x[N-1]$

1) Find $f(x[0], \ldots, x[N-1]; \mu, \sigma^2) =$

$$\frac{1}{(2\pi)^{N/2}(\sigma^2)^{N/2}} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n]-\mu)^2 \right\}$$

2) Find μ, σ^2 that maximize $f(x[0], \ldots, x[N-1]; \mu, \sigma^2)$ given $x[n]$

- Simpler to max $\ln f(\cdot) \Rightarrow \max_{\mu,\sigma^2} -\frac{N}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-\mu)^2$

set $\frac{\partial}{\partial \mu}(\cdot) = 0 \Rightarrow \hat{\mu} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]$

Set $\frac{\partial}{\partial \sigma^2}(\cdot) = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{N}\sum_{n=0}^{N-1}(x[n]-\hat{\mu})^2$

Example 2: Linear Signal Model in Gaussian Noise

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$$\underline{x}_k = \underbrace{\sum_{l=1}^{p} \underline{h}_l \theta_l}_{\underline{s}} + \underline{w}_k = \underline{H}\underline{\theta} + \underline{w}_k \quad , \quad \underline{w}_k \sim N(0, \underline{R}) \text{ ; indep.}$$

$k = 1, 2, \ldots, K$, $\underline{x}_k : N \times 1$

Signal Model - $\underline{h}_l$ are bases, θ_l coefficients

$$s[n] = A\cos(\omega_0 n) + B\sin(\omega_0 n)$$

$$\underset{N\times 1}{\begin{bmatrix} s[0] \\ s[1] \\ \vdots \\ s[N-1] \end{bmatrix}} = \underbrace{\begin{bmatrix} 1 & 0 \\ \cos\omega_0 & \sin\omega_0 \\ \vdots & \vdots \\ \cos(N-1)\omega_0 & \sin(N-1)\omega_0 \end{bmatrix}}_{\underline{H}\ (N\times 2)} \underbrace{\begin{bmatrix} A \\ B \end{bmatrix}}_{\underline{\theta}\ (2\times 1)}$$

1) $$f(\underline{x}_k; \underline{\theta}) = \frac{1}{(2\pi |\underline{R}|)^{N/2}} \exp\left\{-\frac{1}{2}(\underline{x}_k - \underline{H}\underline{\theta})^T \underline{R}^{-1} (\underline{x}_k - \underline{H}\underline{\theta})\right\}$$

$$f(\underline{x}_1, \underline{x}_2, \ldots \underline{x}_K ; \underline{\theta}) = \frac{1}{(2\pi |\underline{R}|)^{\frac{NK}{2}}} \exp\left\{-\frac{1}{2}\sum_{k=1}^{K}(\underline{x}_k - \underline{H}\underline{\theta})^T \underline{R}^{-1} (\underline{x}_k - \underline{H}\underline{\theta})\right\}$$

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2) $\hat{\underline{\theta}}_{MLE} = \arg\max_{\underline{\theta}} \ln\left[f(\underline{x}_1 \cdots \underline{x}_K ; \underline{\theta})\right]$

$$\min_{\underline{\theta}} \sum_{k=1}^{K} (\underline{x}_k - \underline{H}\underline{\theta})^T \underline{R}^{-1} (\underline{x}_k - \underline{H}\underline{\theta})$$

$$\min_{\underline{\theta}} \; K\underline{\theta}^T \underline{H}^T \underline{R}^{-1} \underline{H}\underline{\theta} - \underline{\theta}^T \underline{H}^T \underline{R}^{-1} \sum_{k=1}^{K} \underline{x}_k - \left(\sum_{k=1}^{K} \underline{x}_k\right)^T \underline{R}^{-1} \underline{H}\underline{\theta} + \sum_{k=1}^{K} \underline{x}_k^T \underline{R}^{-1} \underline{x}_k$$

$$\min_{\underline{\theta}} \; \underline{\theta}^T \left[\underline{H}^T \underline{R}^{-1} \underline{H}\right] \underline{\theta} - \underline{\theta}^T \underline{H}^T \underline{R}^{-1} \bar{\underline{x}} - \bar{\underline{x}}^T \underline{R}^{-1} \underline{H}\underline{\theta} + \underline{C} \; ; \; \bar{\underline{x}} = \frac{1}{K} \sum_{k=1}^{K} \underline{x}_k$$

"Complete the square" (write as perfect square in $\underline{\theta}$)

$$\min_{\underline{\theta}} \left(\underline{\theta} - \left[\underline{H}^T \underline{R}^{-1} \underline{H}\right]^{-1} \underline{H}^T \underline{R}^{-1} \bar{\underline{x}}\right)^T \underline{H}^T \underline{R}^{-1} \underline{H} \left(\underline{\theta} - \left[\underline{H}^T \underline{R}^{-1} \underline{H}\right]^{-1} \underline{H}^T \underline{R}^{-1} \bar{\underline{x}}\right) + \underline{C} - \bar{\underline{x}}^T \underline{R}^{-1} \underline{H} \left(\underline{H}^T \underline{R}^{-1} \underline{H}\right)^{-1} \cdot \underline{H}^T \underline{R}^{-1} \bar{\underline{x}}$$

$$\Rightarrow \hat{\underline{\theta}}_{MLE} = \left[\underline{H}^T \underline{R}^{-1} \underline{H}\right]^{-1} \underline{H}^T \underline{R}^{-1} \bar{\underline{x}}$$

Example 3: Geiger Counter

Radioactive source emits α_i particles according to Poisson distribution $P[\alpha_i;\lambda] = e^{-\lambda}\frac{\lambda^{\alpha_i}}{\alpha_i!}$, $\alpha_i \geq 0$

Observe counts $\alpha_1, \alpha_2, \ldots \alpha_N$ (independent), estimate λ

1) $P[\alpha_1, \alpha_2 \ldots, \alpha_N;\lambda] = \prod_{n=1}^{N} \frac{e^{-\lambda}\lambda^{\alpha_n}}{\alpha_n!} = c\, e^{-N\lambda}\lambda^{\sum_{n=1}^{N}\alpha_n}$

2) max w.r.t. λ — take ln, then max

$$\max_{\lambda} \; -N\lambda + \left(\sum_{n=1}^{N}\alpha_n\right)\ln\lambda$$

Set $\frac{\partial}{\partial\lambda}\bullet = 0 \Rightarrow \hat{\lambda}_{MLE} = \frac{1}{N}\sum_{n=1}^{N}\alpha_n$

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